SYSC 5108 Exam Study
Assignment 4, Question 7

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Question

We have the 1D data

$$x = [1, 2, 3, 9, 10].$$

Assume a two-component Gaussian mixture model

$$p(x) = \pi_1 \mathcal{N}(x \mid \mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x \mid \mu_2, \sigma_2^2),$$

with

$$\pi_1 + \pi_2 = 1, \qquad \sigma_1^2 = \sigma_2^2 = 1.$$

The initial parameters are

$$\pi_1 = 0.6, \qquad \pi_2 = 0.4, \qquad \mu_1 = 2.0, \qquad \mu_2 = 9.5.$$

Find the updated parameter values after one EM iteration.

Course and Textbook Cross-References

- **Chapter 7 slides, Gaussian Mixture Model:** the data density is a weighted sum of Gaussian components with mixing coefficients summing to 1.
- **Chapter 7 slides, EM Algorithm:** first compute the hidden-variable responsibilities in the E-step, then update mixture weights and means in the M-step.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, mixture models:** Gaussian mixtures introduce latent component assignments.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, EM discussion:** EM alternates between estimating latent-variable posteriors and maximizing the expected complete-data log-likelihood.

Step 1: Write the Gaussian Density

Because the variance is fixed at 1, each component density is

$$\mathcal{N}(x \mid \mu, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right).$$

In the E-step, the common factor $1/\sqrt{2\pi}$ cancels in the responsibility ratio, so we only need the exponential terms.

Step 2: E-Step Responsibilities

Let γ_{i1} be the responsibility of component 1 for point x_i. Then

$$\gamma_{i1} = \frac{\pi_1 \mathcal{N}(x_i \mid \mu_1, 1)}{\pi_1 \mathcal{N}(x_i \mid \mu_1, 1) + \pi_2 \mathcal{N}(x_i \mid \mu_2, 1)},$$

and

$$\gamma_{i2} = 1 - \gamma_{i1}.$$

Using

$$(\pi_1, \pi_2, \mu_1, \mu_2) = (0.6, 0.4, 2.0, 9.5),$$

we evaluate the responsibilities for the five data points.

For $x_1 = 1$

$$\gamma_{11} = \frac{0.6\, e^{-(1-2)^2/2}}{0.6\, e^{-(1-2)^2/2} + 0.4\, e^{-(1-9.5)^2/2}} = \frac{0.6e^{-0.5}}{0.6e^{-0.5} + 0.4e^{-36.125}} \approx 1.$$

So

$$\gamma_{11} \approx 1, \qquad \gamma_{12} \approx 0.$$

For $x_2 = 2$

$$\gamma_{21} = \frac{0.6\, e^{-(2-2)^2/2}}{0.6\, e^{-(2-2)^2/2} + 0.4\, e^{-(2-9.5)^2/2}} = \frac{0.6}{0.6 + 0.4e^{-28.125}} \approx 1.$$

So

$$\gamma_{21} \approx 1, \qquad \gamma_{22} \approx 0.$$

For $x_3 = 3$

$$\gamma_{31} = \frac{0.6\, e^{-(3-2)^2/2}}{0.6\, e^{-(3-2)^2/2} + 0.4\, e^{-(3-9.5)^2/2}} = \frac{0.6e^{-0.5}}{0.6e^{-0.5} + 0.4e^{-21.125}} \approx 1.$$

So

$$\gamma_{31} \approx 1, \qquad \gamma_{32} \approx 0.$$

For $x_4 = 9$

$$\gamma_{41} = \frac{0.6\, e^{-(9-2)^2/2}}{0.6\, e^{-(9-2)^2/2} + 0.4\, e^{-(9-9.5)^2/2}} = \frac{0.6e^{-24.5}}{0.6e^{-24.5} + 0.4e^{-0.125}} \approx 0.$$

So

$$\gamma_{41} \approx 0, \qquad \gamma_{42} \approx 1.$$

For $x_5 = 10$

$$\gamma_{51} = \frac{0.6\, e^{-(10-2)^2/2}}{0.6\, e^{-(10-2)^2/2} + 0.4\, e^{-(10-9.5)^2/2}} = \frac{0.6e^{-32}}{0.6e^{-32} + 0.4e^{-0.125}} \approx 0.$$

So

$$\gamma_{51} \approx 0, \qquad \gamma_{52} \approx 1.$$

Responsibility Table

To four decimal places:

x_i	γ_{i1}	γ_{i2}
1	1.0000	0.0000
2	1.0000	0.0000
3	1.0000	0.0000
9	0.0000	1.0000
10	0.0000	1.0000

Hence the effective component counts are

$$N_1 = \sum_{i=1}^{5} \gamma_{i1} \approx 3, \qquad N_2 = \sum_{i=1}^{5} \gamma_{i2} \approx 2.$$

Step 3: M-Step Updates

With fixed variance, the updated mixing coefficients and means are

$$\pi_1^{\text{new}} = \frac{N_1}{5}, \qquad \pi_2^{\text{new}} = \frac{N_2}{5},$$

$$\mu_1^{\text{new}} = \frac{\sum_{i=1}^{5} \gamma_{i1} x_i}{N_1}, \qquad \mu_2^{\text{new}} = \frac{\sum_{i=1}^{5} \gamma_{i2} x_i}{N_2}.$$

Update the Mixing Coefficients

$$\pi_1^{\text{new}} = \frac{3}{5} = 0.6, \qquad \pi_2^{\text{new}} = \frac{2}{5} = 0.4.$$

Update the Means

For component 1,

$$\mu_1^{\text{new}} = \frac{1+2+3}{3} = 2.0.$$

For component 2,

$$\mu_2^{\text{new}} = \frac{9+10}{2} = 9.5.$$

Final Answer

After one EM iteration,

$$\boxed{\pi_1^{\text{new}} = 0.6000, \qquad \pi_2^{\text{new}} = 0.4000,}$$

$$\boxed{\mu_1^{\text{new}} = 2.0000, \qquad \mu_2^{\text{new}} = 9.5000.}$$

Why Nothing Changed

The data are already almost perfectly separated into two groups:

$$\{1, 2, 3\} \qquad \text{and} \qquad \{9, 10\}.$$

The initial means 2.0 and 9.5 already sit at the centers of those groups, and the initial weights 0.6 and 0.4 already match the proportions

$$\frac{3}{5} \qquad \text{and} \qquad \frac{2}{5}.$$

So the E-step assigns almost all responsibility to the obvious component for each point, and the M-step returns essentially the same parameters.

Exam Shortcut

For a clearly separated two-component GMM question:

1. Compute the responsibilities γ_{ik}.
2. Sum them to get N_k.
3. Update $\pi_k = N_k/M$.
4. Update each mean by the responsibility-weighted average.

If the clusters are far apart, the responsibilities will be nearly 0 or 1, which makes the arithmetic very fast.

Sources Used

- Assignment 4 PDF, Question 7.
- Local submitted Assignment 4 PDF checked for consistency of the final values.
- Chapter 7 slides on Gaussian mixture models and the EM algorithm.
- Goodfellow, Bengio, and Courville, *Deep Learning*, sections on mixture models and EM.